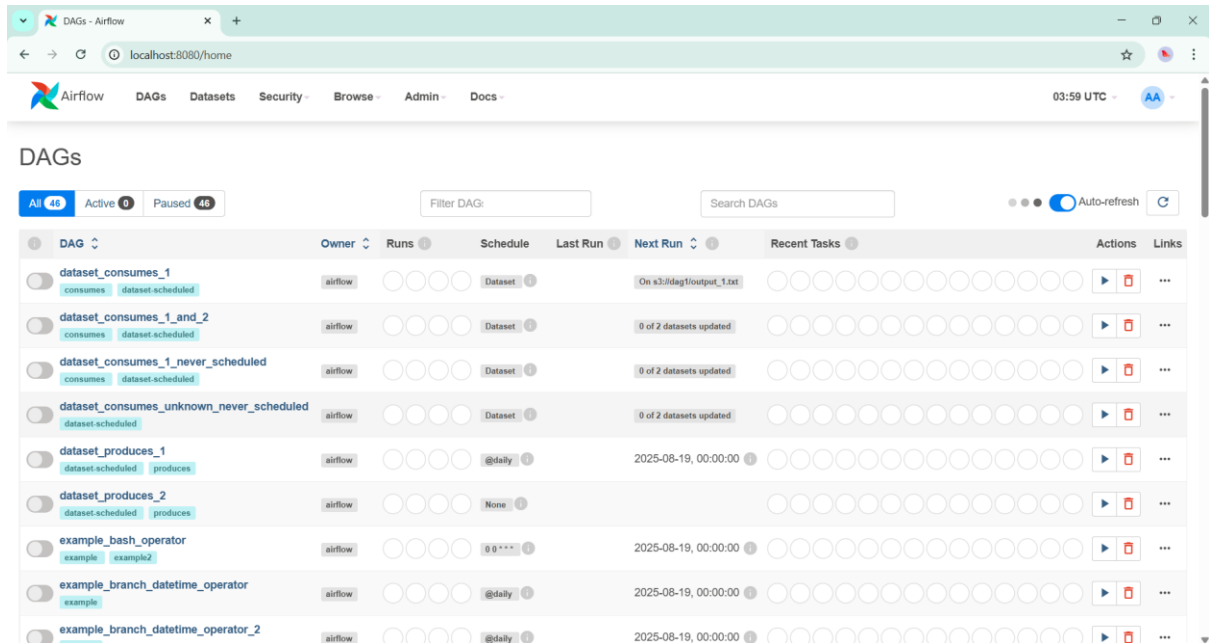


Coding Challenge

Steps to Build a Pipeline in Airflow

Step 1: Open Airflow Web UI

- Navigate to `http://localhost:8080`
- The Airflow home page will open.



Step 2: Create Connection

- Go to **Admin → Connections**.
- Add a new Postgres connection:
 - **Connection ID:** tutorial_pg_conn
 - **Connection Type:** Postgres
 - **Host:** postgres
 - **Database:** airflow
 - **Login:** airflow
 - **Password:** airflow
 - **Port:** 5432
- Save the connection.

DAGs - Airflow

localhost:8080/home

Airflow DAGs Datasets Security Browse Admin Docs 03:59 UTC

DAGs

All 46 Active 0 Paused 46

Search DAGs

Auto-refresh

Variables
Configurations
Connections
Plugins
Providers
Pools
XComs

DAG	Owner	File	Last Run	Next Run	Recent Tasks	Actions	Links
dataset_consumes_1	airflow	dataset-scheduled		On x3:dag1/output_1.txt			
dataset_consumes_1_and_2	airflow	dataset-scheduled		0 of 2 datasets updated			
dataset_consumes_1_never_scheduled	airflow	dataset-scheduled		0 of 2 datasets updated			
dataset_consumes_unknown_never_scheduled	airflow	dataset-scheduled		0 of 2 datasets updated			
dataset_produces_1	airflow	dataset-scheduled produces		2025-08-19, 00:00:00			
dataset_produces_2	airflow	dataset-scheduled produces					
example_bash_operator	airflow	example example2		2025-08-19, 00:00:00			
example_branch_datetime_operator	airflow	example		2025-08-19, 00:00:00			
example_branch_datetime_operator_2	airflow	example		2025-08-19, 00:00:00			

List Connection - Airflow

localhost:8080/connection/list/

Airflow DAGs Datasets Security Browse Admin Docs 03:59 UTC

List Connection

Search

+ Actions

Record Count: 0

No records found

Add Connection - Airflow

localhost:8080/connection/add

Airflow DAGs Datasets Security Browse Admin Docs 04:00 UTC

Add Connection

Connection Id *

Connection Type * Amazon Elastic MapReduce

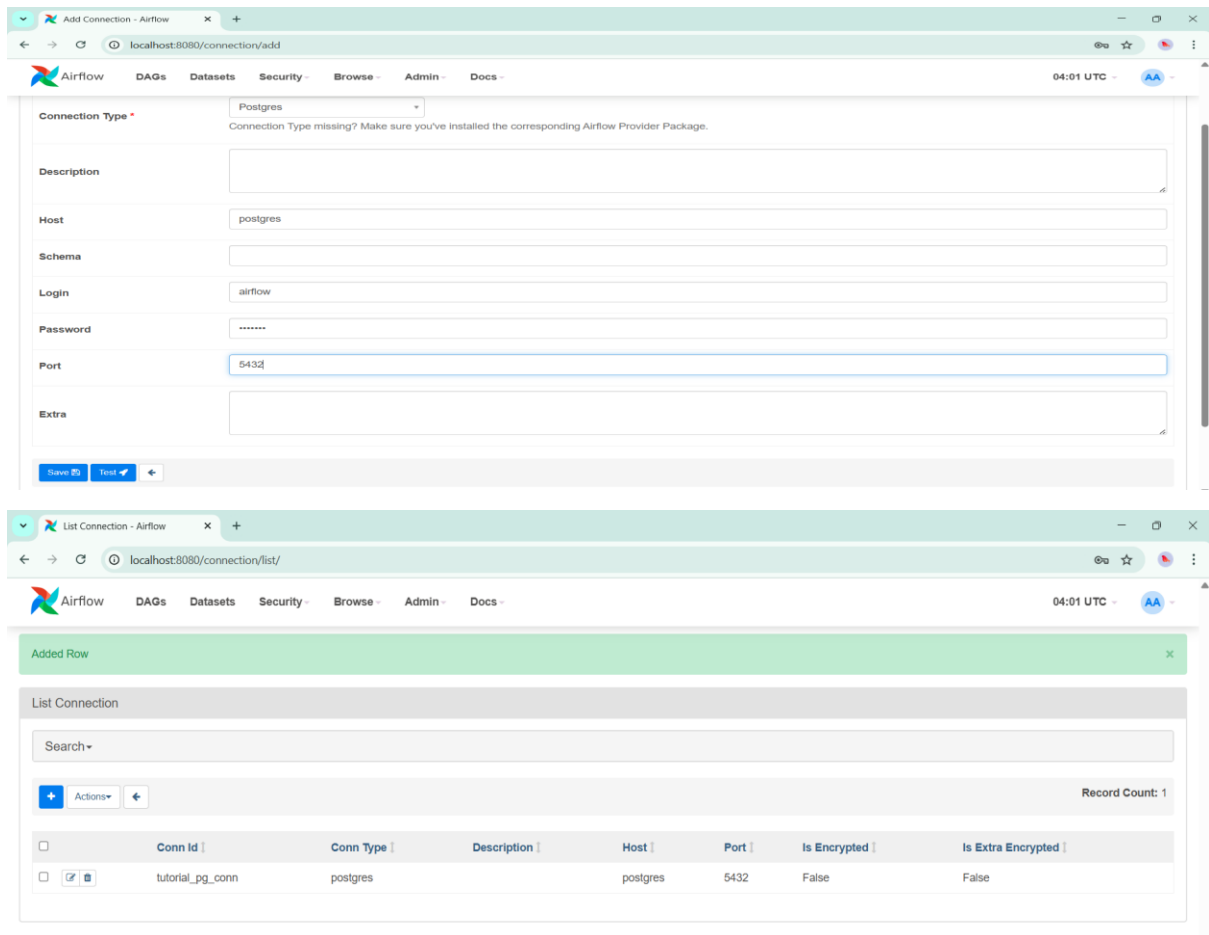
Connection Type missing? Make sure you've installed the corresponding Airflow Provider Package.

Description

Run Job Flow Configuration

```
{
  "Name": "MyClusterName",
  "ReleaseLabel": "emr-5.36.0",
  ...
}
```

Save Test

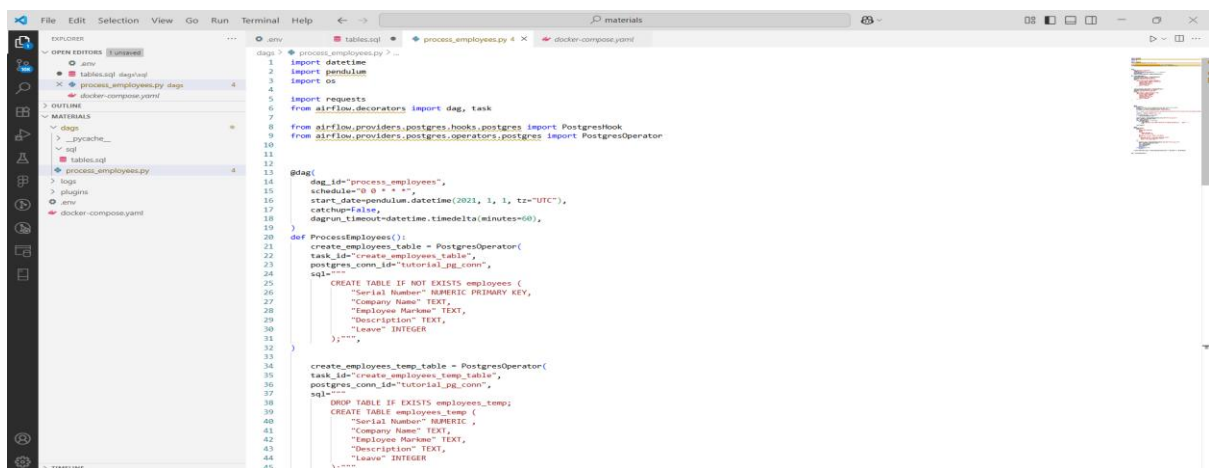


Step 3: Create SQL File

- Go to dags/sql/
- Create tables.sql with SQL queries for creating required tables.

Step 4: Create Python DAG File

- Go to dags/
- Create process_employee.py:

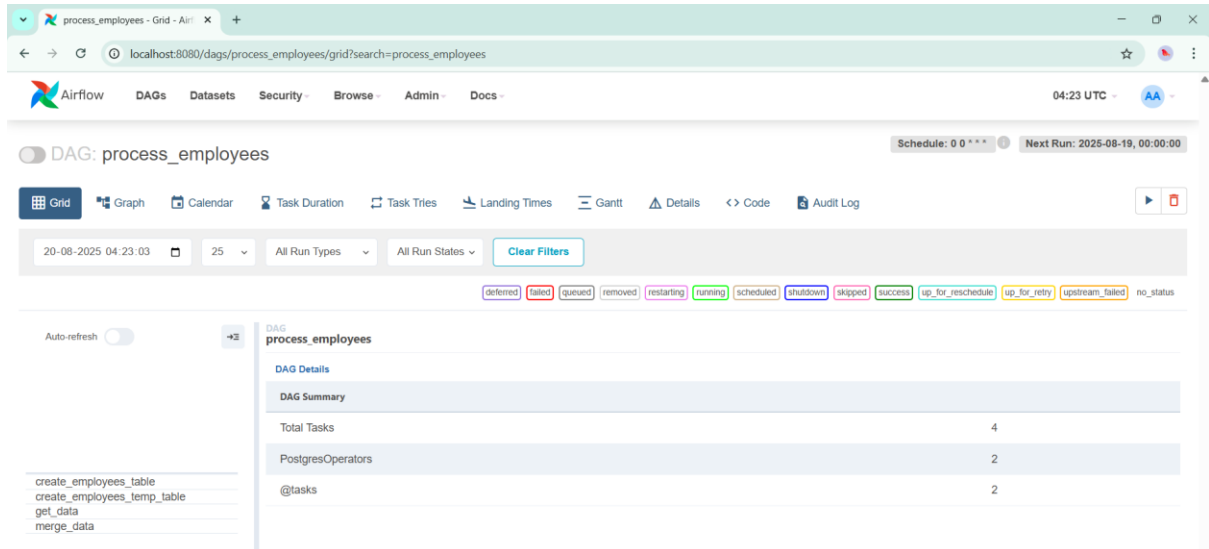


Step 5: Deploy the DAG

- Place process_employee.py in the dags/ directory.
- Airflow will detect it.

Step 6: Run the DAG

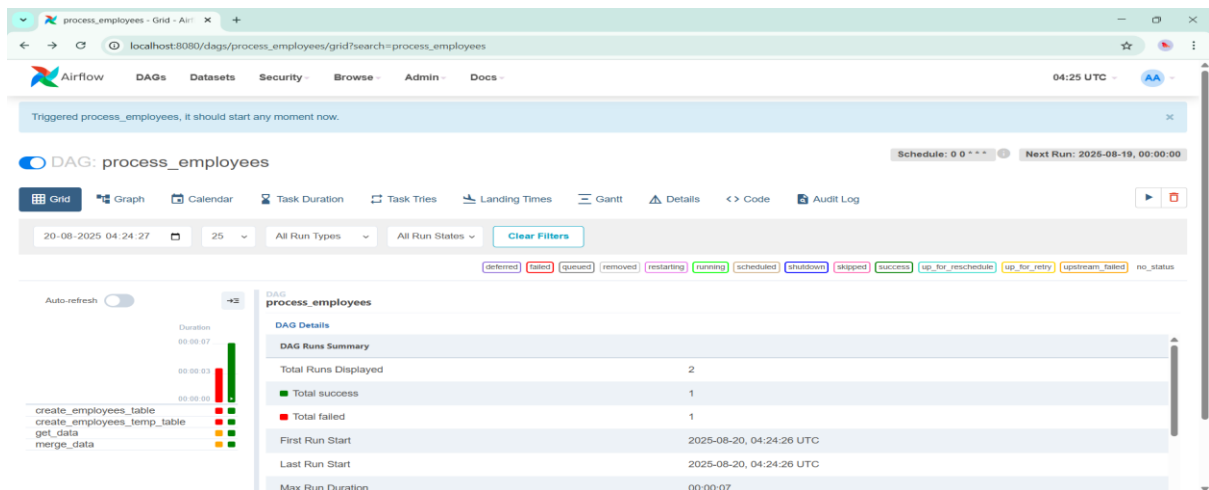
- Trigger the DAG from the web UI.
- Monitor task execution in **Graph View** or **Tree View**.



Viewing Pipelines in Airflow

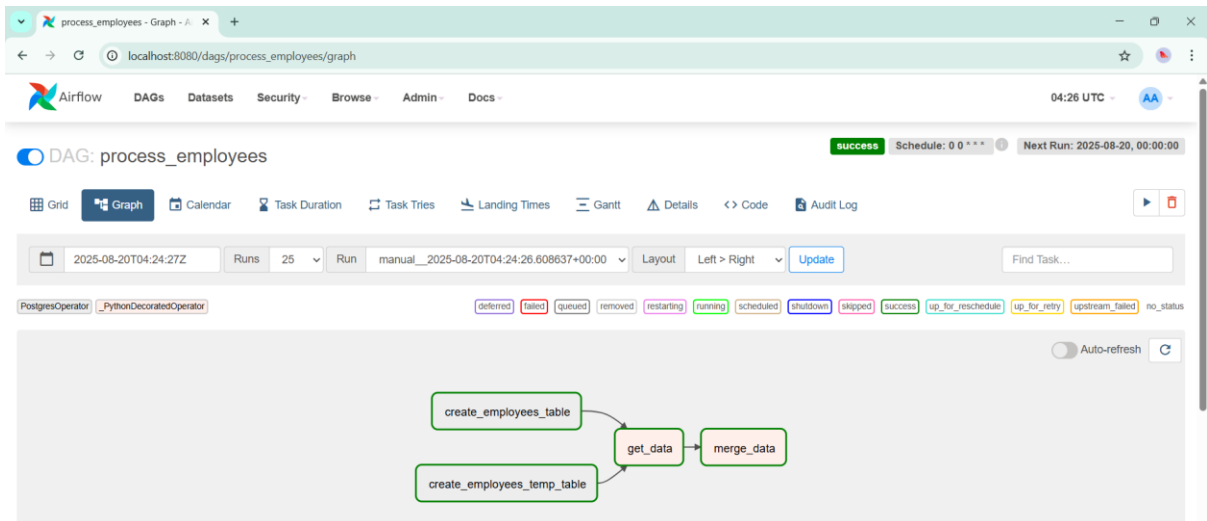
Grid View

- Displays DAG runs and tasks in a timeline-like grid.
- Each column represents a DAG run, and each row represents a task.
- Provides quick insights into task status across multiple runs.



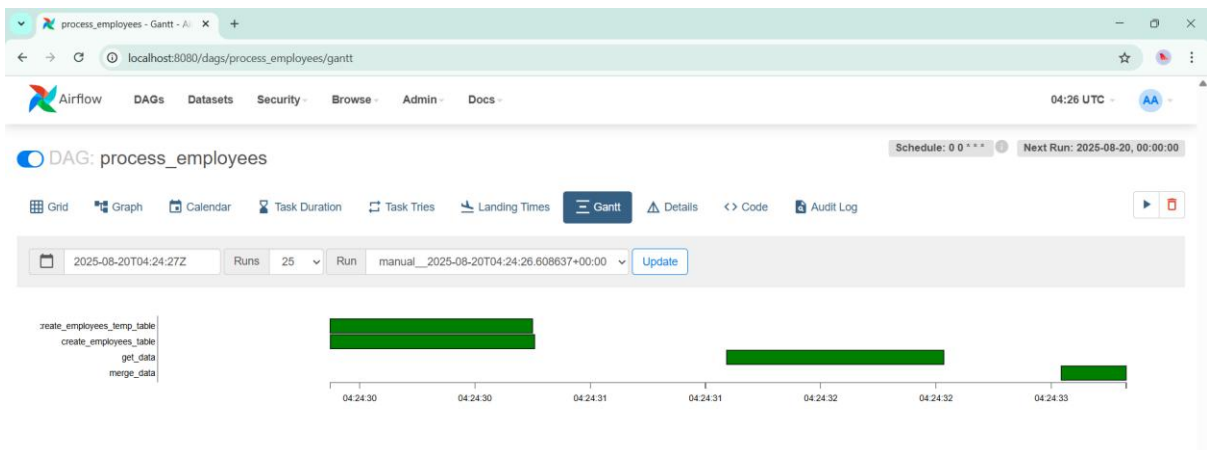
Graph View

- Shows DAG structure as a directed acyclic graph.
- Visualizes task dependencies and execution flow.
- Helpful for understanding pipeline logic.



Gantt View

- Visualizes task execution duration.
- Helps identify bottlenecks and overlapping tasks.
- Useful for performance monitoring.



DAG Details

- Shows overall information about a DAG.
- Includes schedule, description, start date, owners, and configuration.
- Links to logs, task instances, and code view.

process_employees - DAG Details

localhost:8080/dags/process_employees/details

Airflow DAGs Datasets Security Browse Admin Docs 04:28 UTC

DAG: process_employees Schedule: 0 0 * * * Next Run: 2025-08-20, 00:00:00

Grid Graph Calendar Task Duration Task Tries Landing Times Gantt Details Code Audit Log

DAG Details

failed success upstream_failed

Schedule Interval	0 0 * * *
Catchup	False
Started	True
End Date	None
Max Active Runs	0 / 16
Concurrency	16
Default Args	{}
Tasks Count	4
Task IDs	['create_employees_table', 'create_employees_temp_table', 'get_data', 'merge_data']
Relative file location	process_employees.py
Owner	airflow
Owner Links	None

Summary

Using **Docker Compose with YAML**, Apache Airflow can be set up quickly. A sample **Employee Processing Pipeline** demonstrates how to configure connections, create SQL scripts, define DAGs, and execute them. The Airflow UI provides powerful tools like **Grid View, Graph View, Gantt View, and DAG Details** for comprehensive monitoring and debugging.